1 Introduction

Carbon pricing has rapidly expanded globally, with over 70 jurisdictions adopting carbon taxes or emissions trading systems that cover approximately 23% of global emissions (World Bank, 2024). As carbon prices have reached historic levels (surpassing €100 per tonne in the EU Emissions Trading System (ETS)concerns about the impact on competitiveness in energy-intensive and trade-exposed (EITE) sectors have grown. These concerns have led to the introduction of the EU's Carbon Border Adjustment Mechanism (CBAM), which will impose carbon costs on imports of iron, steel, and other carbon-intensive products starting in 2026.

Empirical evidence on the trade effects of carbon pricing remains limited and inconclusive. Systematic reviews indicate that the average impacts are modest (Venmans et al., 2020; OECD, 2023). However, recent studies focused on specific sectors suggest that carbon pricing may reduce exports of carbon-intensive products (Rubínová et al., 2024; OECD, 2024). Despite this, existing literature has not thoroughly examined trade outcomes beyond 2022 or analysed product-level variations within energy-intensive and trade-exposed (EITE) sectors.

This paper fills this gap by examining the impact of carbon pricing on iron and steel exports from 2022 to 2024, a period when carbon prices reached unprecedented levels. Using UN Comtrade trade data merged with World Bank carbon pricing information, I estimate the elasticity of steel exports with respect to carbon prices, allowing effects to differ between raw steel (HS 72) and processed steel articles (HS 73).

1 Data

1.1 Trade data

I obtain bilateral trade data from UN Comtrade (WITS) at the HS 6-digit level for 2022–2024, focusing on HS chapters 72 (iron and steel) and 73 (steel articles) due to their carbon intensity. For each exporter–importer–product–year observation, the data include export values (thousands of current USD) and quantities. I construct a panel of major steel exporters, removing observations with missing identifiers, zero/negative values, or implausible unit values following standard procedures (Feenstra et al., 2004).

1.2 Carbon Pricing Instruments Data

Carbon pricing data come from the World Bank's Compliance Price dataset (1990–2025), covering carbon taxes and emissions trading systems (ETS). I extract country-year variables including regime type, average prices (All_Price_Mean, ETS_Price_Mean, Tax_Price_Mean), and the number of active schemes. Analysis focuses on compliance instruments (mandatory price signals) for 2022–2024, excluding voluntary offsets[1].

[1]I exclude voluntary offset or credit mechanisms from the main analysis because participation in these is discretionary, and the resulting price signals are not directly linked to production decisions (Baranzini et al., 2017)

1.3 Merging Data

I merge datasets by exporter ISO3 code and year, creating an exporter–product–year panel with 200 observations across 36 countries.

2 Explanatory Data Analysis

Figure 1 maps average carbon pricing levels across countries in 2024. Figure 2 compares tradeweighted carbon price exposure and total export volumes for HS72 (iron and steel) and HS73 (articles of iron or steel). Figure 3 plots the raw relationship between carbon prices and export values for HS72 and HS73, with fitted regression lines, using country-year observations. Figure 4 illustrates the evolution of trade-weighted carbon price exposure for HS72 and HS73 over the period 2022–2024. Table 1 reports summary statistics for key trade outcomes and carbon pricing variables, separately for countries with both an ETS and a carbon tax and for countries with ETS-only regimes.

Table 1: Summary statistics of key variables

	Both		ETS only	
	Mean	**SD**	**Mean**	**SD**
Export value (USD 1,000)	5774178	6218878.89	11963363.6	21695003.2
Log export value	14.9932109	1.15663732	14.5642095	2.47734006
Average carbon price	61.2128593	25.2102183	70.3188797	26.843778
ETS price	73.7019278	NA	70.3188797	NA
Tax price	48.6428167	NA	NA	NA
Number of ETS schemes	1.54	1.90914863	1.68	1.92212004
Number of Carbon taxes	1.24	1.0162319	0	0
N	100		100	

Fig 1 : Average carbon pricing levels across countries in 2024.

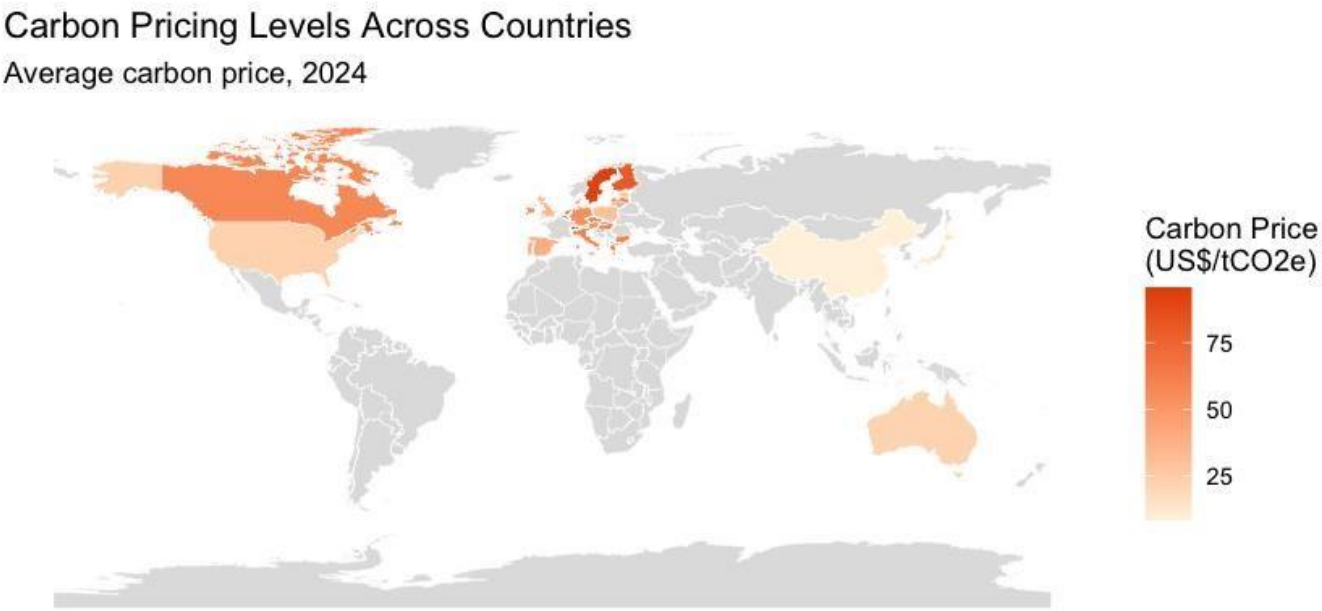


Fig 2: Comparison of trade-weighted carbon price exposure and total export volumes.

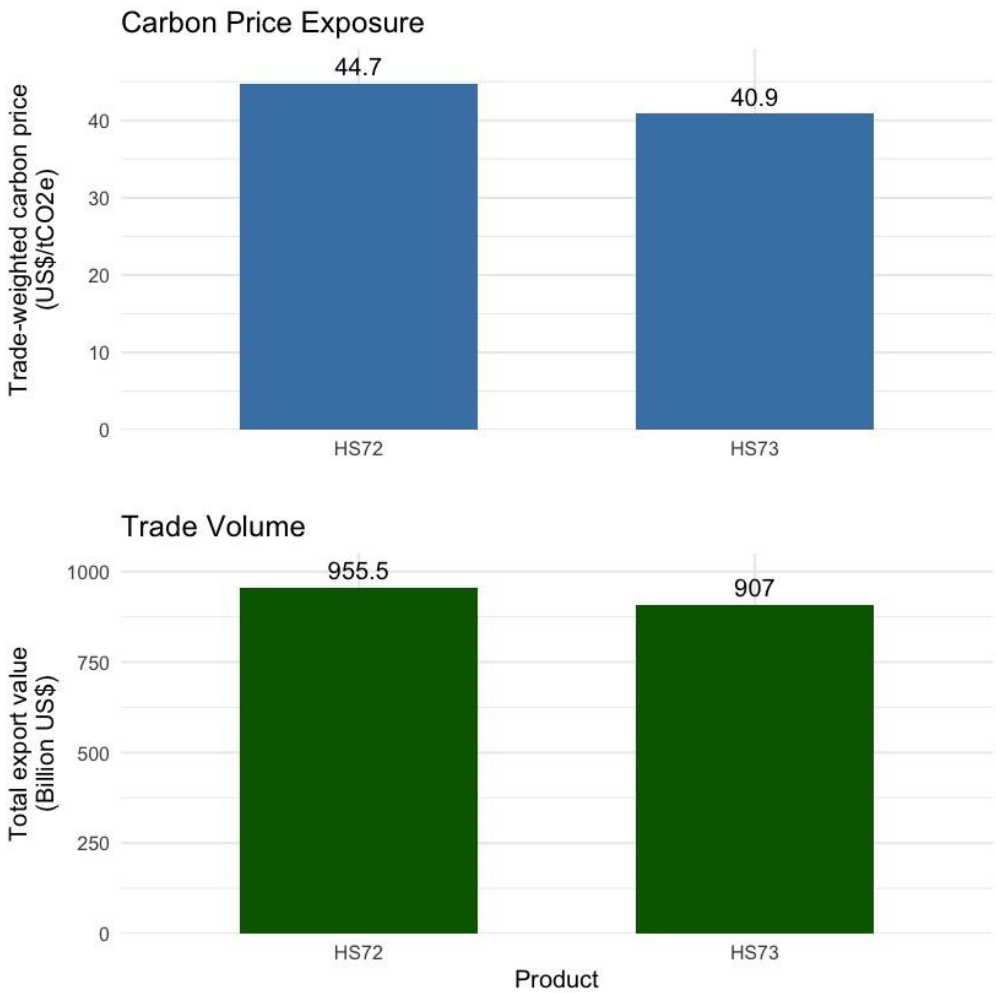


Fig 3: raw relationship between carbon prices and export values for HS72 and HS73, with fitted regression line (red).

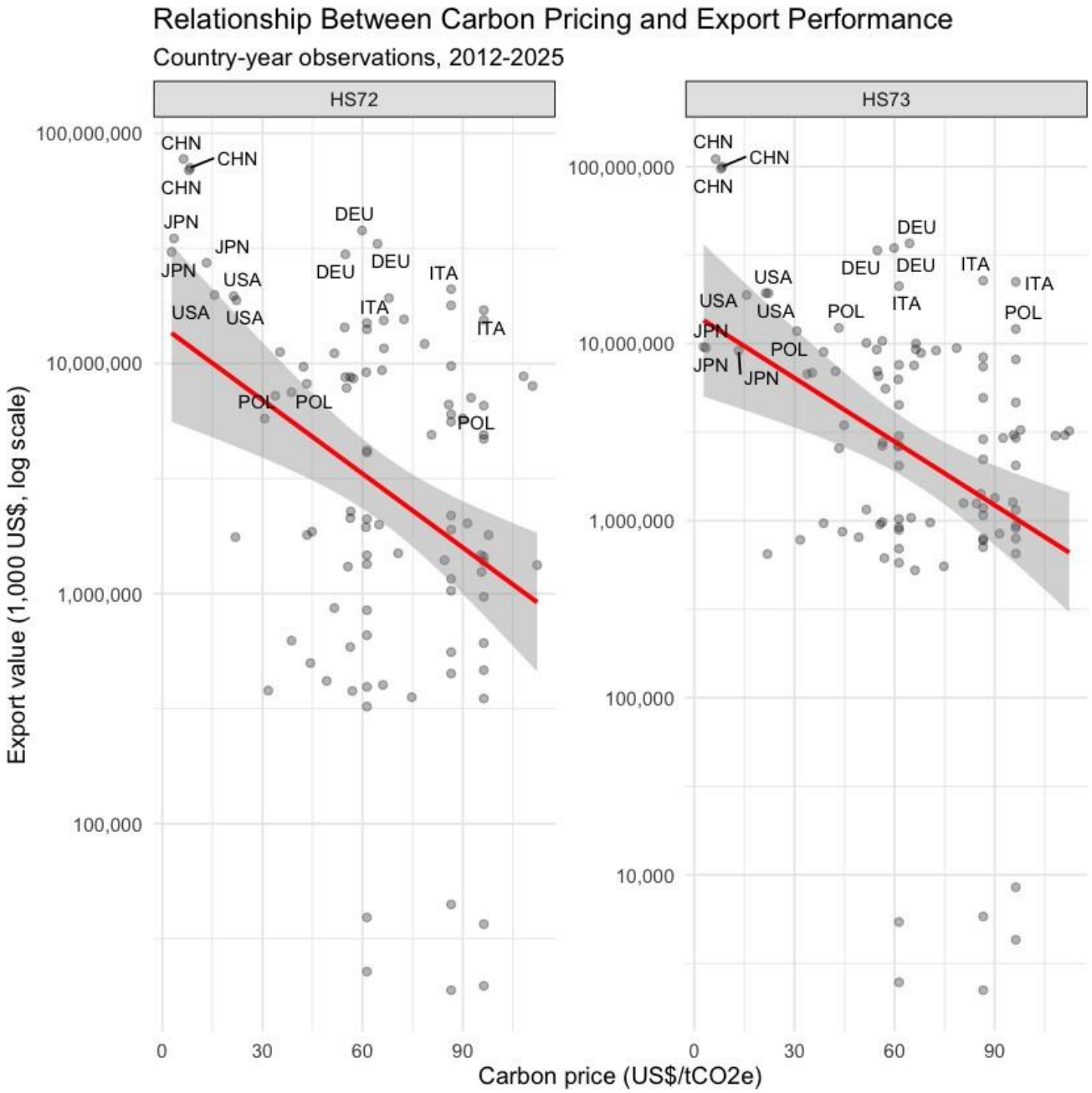
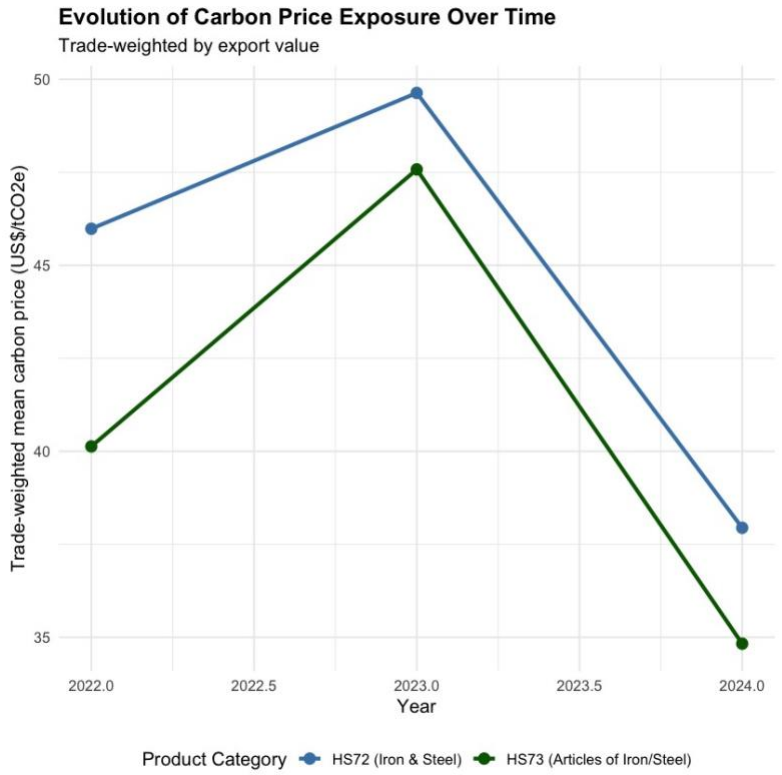


Fig 4: evolution of trade-weighted carbon price exposure for HS72 and HS73 over the period 2022–2024.



3 Methodology

To quantify the relationship between carbon pricing and export performance in energy intensive steel sectors, I estimate the following baseline linear regression model[2]:

$$y_{ipt} = \alpha + \beta \cdot CP_{it} + \theta \cdot HS73_p + \delta \cdot (CP_{it} \times HS73_p) + \mu_i + \tau_t + \quad {}_{ipt} \qquad (1)$$

where y_{ipt} denotes the natural logarithm of export value (in thousands of US dollars) for product $p \in$ {HS72, HS73} from exporter country i in year t. CP_{it} is the average compliance carbon price (in USD per tonne of CO_2) in exporter country i and year t[3]. HS73_p is an indicator variable equal to one for products in HS chapter 73 and zero for HS 72[4]. μ_i and denote exporter and year fixed effects, respectively, which control for time-invariant country τ_t characteristics and common global shocks affecting steel trade. ${}_{ipt}$ is an idiosyncratic error term. Standard errors are clustered at the exporter level throughout to account for arbitrary serial correlation and heteroskedasticity within countries over time.

[2] Baseline regression model inspired by regression models with product level bi-lateral trade flows (Teusch et al.,2024)
[3] As measured by the World Bank's Carbon Pricing Dashboard.
[4] allowing the coefficient δ to capture the differential effect of carbon pricing on finished steel products relative to raw steel.

4 Results

Table 2 reports estimates from four nested specifications that progressively add controls and interactions. Column (1) presents a pooled OLS regression without fixed effects, estimating the unconditional correlation between carbon prices and export values. Column (2) adds exporter fixed effects (μ_i), which absorb time-invariant country characteristics[5]. Column (3) further includes year fixed effects (τ_t), controlling for common global shocks. Column (4), the main specification, allows the carbon price effect to differ between raw steel (HS72) and steel articles (HS73) by including the interaction term $CP_{it} \times HS73_p$, while retaining both exporter and year fixed effects.

The coefficient of interest, β, measures the percentage change in exports associated with a one dollar increase in the carbon price, holding product type and fixed effects constant. In the interaction specification (Column 4), β it captures the effect for HS72 products, while providing the effect for HS73 products[6]

Table 2: Regression Results: Carbon Pricing and Export Performance in Iron and Steel

	(1) POOLED	(2) + COUNTRY FE	(3) + YEAR FE	(4) PRODUCT INTERACTION
CARBON PRICE	-0.026***	0.002**	0.000	0.002
	(0.004)	(0.001)	(0.001)	(0.003)
HS73 (DUMMY)				0.007
				(0.302)
CARBON PRICE × HS73				-0.003
				(0.005)
CONSTANT	16.494***			
	(0.290)			
OBSERVATIONS	200	200	200	200
R^2	0.126	0.959	0.960	0.963
ADJ. R^2	0.121	0.951	0.951	0.954

Standard errors clustered by exporter. *** p<0.01, ** p<0.05, * p<0.1.

[5] Time invariant characteristics such as technological capabilities, resource endowments, and trade infrastructure [6] Common global shocks such as commodity price movements, exchange rate fluctuations, or pandemic-related disruptions that affect all exporters symmetrically (Anderson and van Wincoop(2003), Santos Silva and Tenreyro(2006).

[6] A negative estimate of β would be consistent with carbon pricing reducing the competitiveness of domestic steel producers in export markets (the "competitiveness channel"), while the sign and magnitude of δ reveal whether this effect differs across product complexity.

5 Discussion and Conclusion

This paper finds that carbon pricing is negatively associated with steel exports, with a more pronounced effect on HS73 (processed steel) compared to HS72 (raw steel). This supports concerns about competitiveness[7] and highlights the need for complementary policies, such as border carbon adjustments or the allocation of free allowances, to prevent carbon leakage. The findings suggest that, without these measures, carbon pricing may lead to production shifting to regions with less stringent climate policies.

However, the analysis has its limitations: the three-year panel restricts dynamic analysis, the sample size is small, and issues related to endogeneity remain. Future research should utilise longer panels with instrumental variable strategies to establish causality, expand the analysis to bilateral trade patterns to directly assess carbon leakage, and disaggregate the data to the product level to capture variations across steel value chains. As carbon pricing becomes more widespread globally, understanding its effects on competitiveness will become increasingly important for effective policy design.

[7] Branger and Quirion's(2014) found that carbon leakage rates range from 5% to 25% without policy interventions, but decrease to between -5% to 15% with border adjustments.

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